

1.7 Quiz Solutions: Measurement

Notation: = joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once.

1. Significant digits.

(a) **3**

(c) **2**

(e) **1**

(b) **3**

(d) **4**

(f) **3**

(a) Leading zeros do not count; the trailing zero after the decimal point does. (b) A zero between nonzero digits always counts. (e) 0.000 09 has a single measured digit, the 9.

2. Round to three significant figures.

(a) $\sqrt{13} = 3.6055512 \dots \approx$ **3.61**

(b) 8848.86 m $\approx$ **8850 m**, better written **8.85×10^3 m**

The third figure is the 4 in the tens place; the 8 after it rounds it up.

3. Scientific notation.

(a) 9,460,730,000,000 km = 9.46073×10^{12} km $\approx$ **9.46×10^{12} km**

(b) 0.000 000 000 000 000 000 160 2 C = 1.602×10^{-19} C $\approx$ **1.60×10^{-19} C**

Six groups of three zeros is 18 zeros, so the 1 is in the 19th decimal place: the exponent is -19 . Write 1.60, not 1.6 — the trailing zero is the third significant figure.

4. Ordinary decimal numbers.

(a) 7.0×10^{-5} m = **0.000 070 m**

Keep the trailing zero: 7.0 has two significant figures and so must the decimal form. 0.00007 m loses one.

(b) 5.46×10^7 km = **54,600,000 km**

5. Reading the ruler. The bar ends between the 8.3 and 8.4 cm marks, a little past halfway.

(a) **8.36 cm** — the 8.3 is certain and the final 6 is the estimated digit. *Accept 8.35 to 8.37.*

(b) **Three significant figures.**

(c) $8.36 \text{ cm} \times \frac{10 \text{ mm}}{1 \text{ cm}} =$ **83.6 mm**

6. Convert.

(a) $508 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 1666.24 \text{ ft} \approx$ **1670 ft**, or **1.67×10^3 ft**

The published height is 1667 ft, so this checks.

$$(b) \ 310 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 86.111 \dots \approx \mathbf{86.1 \text{ m/s}}$$

7. Cheetah.

$$29 \frac{\text{m}}{\text{s}} \times \frac{3.28 \text{ ft}}{1 \text{ m}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 64.854 \dots \approx \mathbf{65 \text{ mi/h}}$$

Two significant figures, set by the 29 m/s. Accept 64.9 mi/h for full credit on the conversion; the sig-fig point is worth discussing when the quiz is returned.

Early finisher

Two steps: (1) the rate relationship, rate $\times$ time = quantity or a rearrangement; (2) convert that result to the units asked for, carrying the unrounded value. Convert first when it simplifies Step 1; metric changes are done in the head.

8. Radio command to Mars. $t = \frac{d}{r}$; d is in km, so first $3.00 \times 10^8 \text{ m/s} = 3.00 \times 10^5 \text{ km/s}$.

$$\text{Step 1: } t = \frac{5.46 \times 10^7 \text{ km}}{3.00 \times 10^5 \text{ km/s}} = 182 \text{ s}$$

$$\text{Step 2: } 182 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 3.0333 \dots \approx \mathbf{3.03 \text{ min}}$$

That is at closest approach, and up to about 24 minutes when Mars is far — which is why a Mars rover cannot be driven with a joystick.

9. Widening of the Atlantic. $t = 2026 - 1492 = 534 \text{ yr}$.

$$\text{Step 1: } d = r \cdot t = 2.5 \frac{\text{cm}}{\text{yr}} \times 534 \text{ yr} = 1335 \text{ cm}$$

$$\text{Step 2: } 1335 \text{ cm} = 13.35 \text{ m} \approx \mathbf{13 \text{ m}}$$

Two significant figures, set by “about 2.5 cm”; the 534 years is a count and does not limit the answer. About the length of a city bus — in five centuries.

10. Shower.

$$\text{Step 1: } \text{quantity} = r \cdot t = 2.5 \frac{\text{gal}}{\text{min}} \times 8.0 \text{ min} = 20 \text{ gal}$$

$$\text{Step 2: } 20 \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 75.7 \text{ L} \approx \mathbf{76 \text{ L}}$$

Two significant figures: both 2.5 gal/min and 8.0 min have two.

11. Water per person per day.

$$\text{Step 1: } r = \frac{1.0 \times 10^9 \text{ gal/day}}{8.5 \times 10^6 \text{ people}} = 117.647 \dots \text{ gal per person per day}$$

$$\text{Step 2: } 117.647 \dots \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 445.29 \dots \text{ L} \approx \mathbf{4.5 \times 10^2 \text{ L}} \text{ (450 L)}$$

Both data values have two significant figures, so the answer gets two; 445 L overstates the precision. Full credit for the calculation, the sig-fig mark only for two.